

Qn	Answer	Explanation
1	A	Surface area of cylinder $A = 2\pi rL = 282.7 \text{ cm}^2$ $\frac{\Delta A}{A} = \frac{\Delta r}{r} + \frac{\Delta L}{L}$ $\Delta A = \left(\frac{\Delta r}{r} + \frac{\Delta L}{L} \right) A = \left(\frac{0.1}{3.0} + \frac{0.1}{15.0} \right) 282.7 = 10 \text{ cm}^2 \text{ (to 1 s.f.)}$ Therefore $A = (280 \pm 10) \text{ cm}^2$
2	C	During free fall